

CHE301 HW7

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Problem A - Max Amount of Water in a Gas Stream

For problem A, we are given a dry gas with a known composition. We want to know how much water that can be held by the dry gas. In other words, how much water can the gas hold until a drop of liquid is formed.

To solve this problem, we utilize an equilibrium condition for water. This condition states that the fugacity of the water within the gas mixture is equal to the fugacity of the water in a pure drop of liquid water. Both of these parameters depend on pressure and temperature, and the mole fraction of the water is taken into account. See the equation below.

$$f_w^{\text{gas}}(T, P, y_w) = f_w^{\text{pure}}(T, P)$$

In order to solve for the mole fraction of water, we will make an initial guess and eventually use goal-seeking to satisfy the above equation.

Let us begin by converting all the given weight fractions into molar fractions. To do this, we choose a basis to represent the total mass of the dry gas. The basis will be 100 grams.

$$m_{\text{methane}} = 100 \text{ g} \times 0.60487 = 60.487 \text{ g}$$

$$n_{\text{methane}} = 60.487 \text{ g} \times \frac{1 \text{ mol}}{16.043 \text{ g}} = 3.770 \text{ moles}$$

Repeat to get the amount of moles for every component. The following table results.

Component	Moles
Methane	3.770
Ethane	0.474
Propane	0.190
Isobutane	0.142
n-butane	0.142

Add all the moles, then divide the number of moles of a component by the total number of moles. This yields the following table of mole fractions.

Component	Mole fraction
Methane	0.799
Ethane	0.101
Propane	0.040
Isobutane	0.030
n-butane	0.030

Now, let us convert the given pressure and temperature into units of bar and degrees celsius. We input the given values in to the functions, **PSIAtoBAR** and **FtoC**. The conversion gives us the following values.

$$\text{Temperature} = 18.3333 \text{ }^{\circ}\text{C}$$

$$\text{Pressure} = 62.0526 \text{ bar}$$

Inputting these values into the VLE calculations section of the CHE calculator gives the following.

Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10
Molar Composition	1.0000	0.799000	0.101000	0.040000	0.030000	0.030000
Temperature, C	18.3333					
Pressure, bar	62.0526					

At this point, we should make an initial guess for the mole fraction of water. This allows us to rewrite the molar compositions of each component as a function of the mole fraction of water. We use the equation below.

$$z_{ij} = (1 - z_w) \times z_i$$

Where z_i represents the mole fraction of a component i without any water, z_w represents the mole fraction of water, and z_{ij} represents the newly calculated mole fraction of component i in the presence of water.

Now, let us make an initial guess for z_w . We selected a dimensionless value of 0.001, since water has a very low volatility in comparison to the natural gasses in the dry gas stream. This means that only a very small amount of water will vaporize under the given conditions. Let us add these adjustments to the calculator, shown below.

Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O
Molar Composition	1.0000	0.799200	0.099900	0.039960	0.029970	0.029970	0.001000
Temperature, C	18.3333						
Pressure, bar	62.0526						

Let us move on to calculating the fugacity of the water in the dry gas. To do this, we need the fugacity coefficient, which in turn requires the compressibility factor.

To calculate the compressibility factor, Z , we have to assume that the composition of the vapor is the same as the overall composition. We are given the temperature and pressure, so from here we can utilize the function **ZComp** in the calculator. The function arguments are shown below.

Function Arguments
?
X

ZComp

MolFrac
yi
= {0.7992,0.0999,0.03996,0.02997,0.02997,0.001}

T
Temp
= 18.33333333

P
Pres
= 62.05259935

Zo
1
= 1

Maxite
= 0.767389895

Calculates the Compressibility Factor.

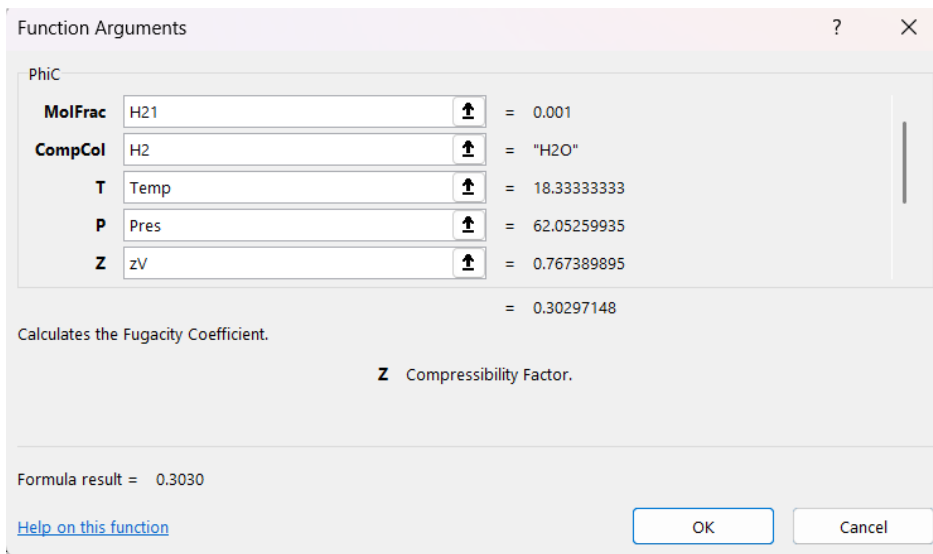
Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.7674

[Help on this function](#)
OK
Cancel

To ensure we calculate the root representing the compressibility for a vapor, we make a guess of 1. This outputs a compressibility factor of 0.7674. From here, we can calculate the fugacity coefficients of each component using the built-in function **PhiC**. However, we are only interested in the fugacity coefficient of water in the mixture. The function arguments are shown below.

This outputs a fugacity coefficient of 0.3030. This value, along with the compressibility factor will change as we iterate a value for z_w .



The image shows a 'Function Arguments' dialog box for the **PhiC** function. The arguments are as follows:

Argument	Value	Result
MolFrac	H21	= 0.001
CompCol	H2	= "H2O"
T	Temp	= 18.33333333
P	Pres	= 62.05259935
Z	zV	= 0.767389895
		= 0.30297148

Calculates the Fugacity Coefficient.

Z Compressibility Factor.

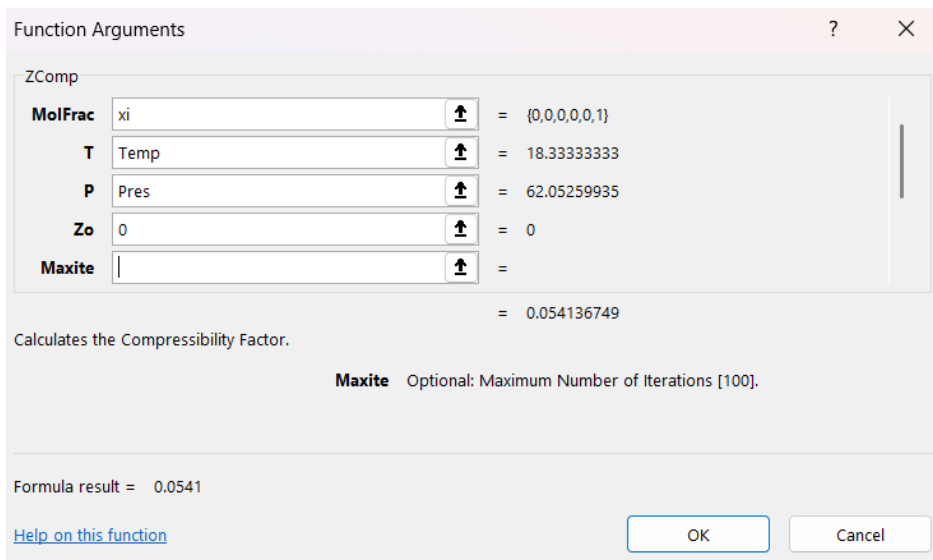
Formula result = 0.3030

[Help on this function](#) [OK] [Cancel]

Now, we simply calculate the fugacity of the water in the mixture by multiplying the fugacity coefficient, pressure, and mole fraction together. This is shown below.

$$f_w^{\text{gas}} = z_w \phi_w^{\text{gas}} P = (0.001)(0.3030)(62.0526 \text{ bar}) = 0.0188 \text{ bar}$$

Next, we calculate the fugacity of the pure liquid water. Again, we can use the function, **ZComp**. The input composition only consists of a mole fraction equal to one for water. The function arguments are shown below.



The image shows a 'Function Arguments' dialog box for the **ZComp** function. The arguments are as follows:

Argument	Value	Result
MolFrac	xi	= {0,0,0,0,1}
T	Temp	= 18.33333333
P	Pres	= 62.05259935
Zo	0	= 0
Maxite		= 0.054136749

Calculates the Compressibility Factor.

Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.0541

[Help on this function](#) [OK] [Cancel]

Use this value of the compressibility factor, 0.0541, to calculate the fugacity coefficient. The function arguments are very similar to the prior calculation, however, we instead use the liquid compressibility factor and the liquid mole fraction rather than their vapor counterparts. The result is 0.0003.

Multiply the fugacity coefficient by the pressure to yield the fugacity of the pure liquid. We can omit the mole fraction, since it is a pure fluid. The calculation is shown below.

$$f_w^{\text{pure}} = \phi_w^{\text{pure}} P = (0.0003)(62.0526 \text{ bar}) = 0.0184 \text{ bar}$$

Since the two calculated fugacities are not equal, we must keep making new guesses until they are. Here, we do this by using goal-seek. Specifically, we change the containing the molar fraction for water until the below equation is satisfied.

$$f_w^{\text{gas}} - f_w^{\text{pure}} = 0$$

The resulting value is **0.000981**. This makes physical sense, as it is very small compared to the hydrocarbon gasses in the mixture that tend to volatilize much more easily. To convert to lb/MMSCF, we perform the following calculations.

First, let's get the moles of dry gas in 1 MMSCF at standard conditions.

$$n_{\text{dry, std}} = \frac{14.7 \times 10^6}{10.7316 \times 520} = 2636.7 \text{ lbmol dry gas}$$

Then, convert the mole fraction water into moles of water per moles of dry gas.

$$y_w = \frac{n_w}{n_{\text{dry}} + n_w}$$

$$\frac{n_w}{n_{\text{dry}}} = \frac{y_w}{1 - y_w}$$

$$\frac{n_w}{n_{\text{dry}}} = \frac{0.000981}{1 - 0.000981} = 0.000982$$

Now, we get the amount of moles of water in 1 MMSCF, which we can then multiply by the molar mass of water to get the pounds per MMSCF.

$$n_w = 0.000982 \times 2636.7 = 2.590 \text{ lbmol water}$$

$$m_w = n_w \times MW_{\text{H}_2\text{O}}$$

$$m_w = 2.590 \times 18.015 = \boxed{46.6 \frac{\text{lb H}_2\text{O}}{\text{MMSCF}}}$$

Problem A - Bubble and Dew Point Temperatures

To calculate the dew point temperature, we will follow a procedure outlined in the textbook. First, we need to make an initial guess for the dew point temperature. To do this, we assume ideal conditions. As a result, we can utilize the following equation.

$$\sum_i \frac{\bar{z}_i}{K_i} = 1, \quad \text{where } K_i = \frac{P_i^{\text{sat}}(T)}{P}$$

Then, we make a random guess for the dew point that we aim to refine. Using this guess, we use either **Psat** or **WilsonPsat** to calculate the saturated pressures at the guessed temperature. Only use **WilsonPsat** if you are dealing with supercritical conditions. Then, use **goalseek** to keep changing the dew point guess until the below equation is satisfied.

$$\sum_i \frac{z_i}{K_i} = 1$$

Performing this procedure gives the following initial guess.

$$T_{\text{dew, guess}} \approx 36.5^\circ\text{C} \text{ (98}^\circ\text{F)}$$

Now, let us set the vapor molar fractions equal to the overall molar fractions, since we are at the dew point. For the liquid molar fractions, we must make another guess. Here, we again use **WilsonPsat** and **Psat** to calculate saturated pressures. We can then use these saturated pressures to calculate initial K values.

$$K^{(o)} = \frac{P_i^{\text{sat}}}{P}$$

Then, we divide the overall molar compositions by these initial K values. Normalization of these results gives us the ballpark for the initial guesses for the liquid mole fractions.

Below, we show values in the following order: saturated pressures based on an initial guess of 35.6 °C, initial K-values, resulting liquid mole fractions before normalization, and resulting liquid mole fractions after normalization.

365.4834	52.0671	84.9528	2.9047	3.3801	0.0501
5.8899	0.8391	1.3690	0.0468	0.0545	0.0008
0.035783	0.063876	0.012653	1.455070	2.102445	0.315133
0.008983	0.016000	0.003075	0.365225	0.527000	0.079762

Now, we can calculate the compressibility factors of both the vapor and liquid phases using **ZComp**. For this first iteration, the values come out to the following:

$$Z_V = 0.8128$$

$$Z_L = 0.2185$$

With these values, we can use **PhiC** to calculate all the corresponding fugacity coefficients. Then, we calculate the K-values:

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}$$

Now that we have K-values, we can set $\alpha = 1$, and get an output for the Rachford-Rice equation. In this first iteration, the Rachford-Rice equation is equal to 0.1825.

Now, we can use goalseek to alter our initial guess for the dew point temperature and set the Rachford-Rice equation equal to zero. From here, the next step is to calculate new values for the liquid mole fractions. We check to see if they are equal to our initial guess, and if not, we perform direct substitution by replacing our initial guess with the newly calculated values.

From there, we can continue to goalseek the Rachford-Rice equation till all the necessary values are obtained. This can be done quickly by selecting the bubble point temperature button in the Excel ribbon. Following this procedure yields the values below.

$$T_{\text{Dew}} = 17.9520^{\circ}\text{C} = 64.3137^{\circ}\text{F}$$

The calculation of the bubble point temperature is very similar. There are a few key differences, but they are all intuitive. We list them below.

- The liquid mole fraction equals the overall ($x_i = z_i$)
- The vapor fraction is equal to zero ($\alpha = 0$)
- The end result should be far lower than the dew point temperature

Again, we start with an initial guess. We use the same method as before, but with a different equation to satisfy. It is shown below.

$$\sum_i z_i K_i = 1$$

With this procedure, the initial bubble point we end up with is -150 °C. By following the procedure we outlined before and taking note of the key differences we listed, calculating the bubble point temperature becomes trivial. As such, we shall omit the explanation, since it would just repeat what we stated earlier. The final value is shown below.

$$T_{\text{Dew}} = 17.9520^{\circ}\text{C} = -67.4036^{\circ}\text{F}$$

Problem B - Degrees of Freedom

The Gibbs phase rule is shown below.

$$F = C - P + 2$$

Where, F represents the degrees of freedom, C represents the number of chemical components, and P represents the number of phases.

There are five components present: nitrogen, oxygen, argon, carbon dioxide, and water. So, C = 5.

Two phases are present in this case, so P = 2.

$$F = C - P + 2 = 5 - 2 + 2 = 5$$

Before counting constraints, there are five degrees of freedom. Let us analyze the problem for constraints and list them out.

- Pressure is given (1)
- Temperature is given (1)
- Dry air composition of four components is fully known (3)

In total, there are five constraints

$$5 - 5 = 0$$

Since there are zero degrees of freedom, this problem is well-posed.

$$\boxed{F = 0}$$

Problem B - CO2 Solubility in Water

As for most problems, we need to start with a guess. In this case, we must make a guess for the mole fraction of water. This allows to calculate respective values for the mole fractions of the dry air in the presence of water. The equation below shows how the mole fractions of the dry air account for the water.

$$z_{ij} = z_i \times (1 - z_{H_2O})$$

Where z_{ij} is equal to the newly calculated mole fraction in the presence of water, z_i is the original mole fraction, and z_{H_2O} is the mole fraction of water. Let's take an initial guess to be **0.05**. We choose a very low mole fraction, since the other components in the mixture are much more likely to volatilize.

Inputting known values for temperature, pressure, and mole fractions yields the following.

Fluid Conditions	Mixture	N2	O2	Ar	CO2	H2O
Molar Composition	1.0000	0.741798	0.198987	0.008816	0.000399	0.050000
Temperature, C	20.0000					
Pressure, bar	2.0000					

Let us calculate the compressibility factor of the vapor mixture. This will allow us to calculate the fugacity coefficient, leading to the fugacity, and eventually to the mole fractions that we need. Again, we shall use **ZComp** to make this calculation. Before we do this, we set the vapor mole fractions equal to the overall mole fractions. To get the fugacity coefficients, we use **PhiC**. These values are all shown in the table below.

Vapor Phase	Mixture	N2	O2	Ar	CO2	H2O
Molar Composition	1.0000	0.741798	0.198987	0.008816	0.000399	0.050000
New Molar Composition						
Compressibility Factor	0.9980					
Fugacity Coefficient		0.9990	0.9980	0.9980	0.9886	0.9686

Our desired fugacities are calculated via their fugacity coefficients, overall pressure, and vapor mole fractions.

$$f_i = \phi_i P y_i$$

$$f_{CO_2} = (0.9886)(2.0 \text{ bar})(0.000399) = 0.0008 \text{ bar}$$

$$f_{H_2O} = (0.9686)(2 \text{ bar})(0.050) = 0.0969 \text{ bar}$$

Now, before we can calculate the mole fractions, we have to determine the Henry's constant for CO₂, since it is under supercritical conditions. We also need to find the saturated pressure of water, which will be done by using steam tables.

For the Henry's constants, we utilize the table below.

	A	B	C	D
OXYGEN	144.4080745	-7775.06	-18.3974	-0.00944354
NITROGEN	164.9940745	-8432.77	-21.558	-0.00843624
ARGON	169.4780745	-8137.13	-23.2547	0.00306357
HELIUM	117.3450745	-4397.06	-16.2365	0.00602486
HYDROGEN	180.0660745	-6993.51	-26.3119	0.0150431
CO	171.7750745	-8296.75	-23.3372	0
CO2	159.8650745	-8741.55	-21.669	0.00110259
H2S	149.5490745	-8226.5	-20.2307	-0.00129405

We can calculate H_{CO_2} via the constant values from the table and the equation below.

$$\ln H = A + \frac{B}{T} + C \ln T + DT$$

We input the given temperature with units of Kelvin, and the following result for $\ln(H)$ is given below.

$$\ln(H) = 7.2742$$

$$H = \exp(7.2742) = 1442.5566$$

To get the saturated pressure of water at the given temperature, we can use the function **WaterPsat**.

$$\mathbf{WaterPsat}(T = 20^\circ\text{C}) \rightarrow 0.0234 \text{ bar}$$

Now, we can calculate mole fractions.

$$x_{CO2} = \frac{0.0008}{1442.5566} = 5.546 \times 10^{-7}$$

$$x_{H2O} = \frac{0.0969}{0.0234} = 4.14$$

Obviously, these values do not add up to 1, so we proceed by iterating via goalseek. The below value is the result.

$$x_{H2O} = 0.999$$

Now we can calculate the ratio of the liquid molar fractions, and then convert them into milligrams of CO₂ per liter of water.

$$\frac{x_{CO2}}{x_{H2O}} = 5.69 \times 10^{-7}$$

$$5.69 \times 10^{-7} \times \frac{MW_{CO_2}}{MW_{H_2O}} \times 10^6 = \boxed{1.3893 \frac{\text{mg CO}_2}{\text{L H}_2\text{O}}}$$

Problem C - Part 1

We want to find the inlet and outlet temperatures of the mixed refrigerant. In the problem statement, we are told that the feed mixed refrigerant is a saturated liquid. From this, we can recognize that the mixture is at its bubble point temperature.

Let us calculate this bubble point temperature. We start with an initial guess, formulated in a similar way from problem A. The equation for the guess is shown below.

$$T_{\text{BUBBLE}}^0 = 0.7 \times \sum_j z_j T_{cj} \approx -81^\circ\text{C}$$

Now, let us set the liquid composition equal to the overall composition. Then, let us make guesses for the vapor composition based on the idea that lighter components will be more prominent in the vapor phase. These guesses are shown below.

N2	CH4	C2H6	C3H8	C4H10
0.396000	0.495000	0.099100	0.000000	0.009900

Now use **Zcomp** and **PhiC** to calculate the compressibility factors and fugacity coefficients, just as in problem A.

With fugacity coefficients calculated, we can get K-values. We use the following equation.

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}$$

Then, we set the vapor fraction equal to zero and use goal seek to manipulate the bubble point temperature until the Rachford-Rice equation is satisfied.

To check our guesses for the vapor composition, we calculate new values for y_i . We use the following equation.

$$y_i = K_i x_i$$

If the new values are not equal to the initial guesses, we perform direct substitution until they are. By using the **Bubble Point Temperature** macro, we can skip manual iteration. The resulting value for the bubble point temperature is given below.

$$T_{\text{BUBBLE}}^0 = -5.8730^\circ\text{C}$$

The final values for each calculated parameter is shown below from a screenshot of the CHE calculator.

Fluid Conditions	Mixture	N2	CH4	C2H6	C4H10
Molar Composition	1.0000	0.150000	0.300000	0.300000	0.250000
Temperature, C					
Pressure, bar	60.0000				

Bubble Point Temperature, C	-5.8730				
Dew Point Temperature, C					
K-Values		2.0289	1.4890	0.6430	0.2243

Mixture	N2	CH4	C2H6	C4H10
1.0000	0.304335	0.446686	0.192911	0.056067
1.0000	0.304335	0.446686	0.192911	0.056068
0.7429				
	0.9877	0.8353	0.4815	0.1054

Mixture	N2	CH4	C2H6	C4H10
1.0000	0.150000	0.300000	0.300000	0.250000
0.2118				
	2.0039	1.2437	0.3096	0.0236

Now that we have the inlet temperature, let us proceed by calculating the inlet enthalpies and entropies. We calculate the inlet enthalpy by using the equation below.

$$h_{\text{mix}}(T, P, \mathbf{z}) = h_{\text{mix}}^{\text{IG}}(T) + h_{\text{mix}}^{\text{R}}(T, P, \mathbf{z})$$

Both of the terms on the right are summations of the value for each component in the mixture. To calculate the enthalpy of the mixture under ideal gas assumptions, we use the calculator function **HIG**. We also multiply by the overall molar composition and then sum it.

For the residual enthalpy, we use the function **HR**. It already factors in the mole fraction so we do not have to multiply it, we can just go ahead and sum it. The values are shown below.

	N2	CH4	C2H6	C4H10	sum
Hig, kj/kmol	-135.1950	-22791.8424	-25889.1371	-32285.2193	-81101.3937
HR, kj/kmol	-3482.2390	-4949.7268	-10464.9755	-24708.3628	-43605.3041
h_mix, kj/kmol	-124706.6978				

This value for the inlet enthalpy will be useful later, when calculating the work produced. Now, let's calculate the inlet entropy. It follows the same pattern. Instead of **HIG** and **HR**, we will now use **SIG** and **SR**

$$s_{\text{mix}}(T, P, \mathbf{z}) = s_{\text{mix}}^{\text{IG}}(T) + s_{\text{mix}}^{\text{R}}(T, P, \mathbf{z})$$

The following table is produced.

	N2	CH4	C2H6	C4H10	sum
Sig, kj/kmol/K	-37.1241	-118.3275	-213.1457	-410.0335	-778.6308
SR, kj/kmol/K	-18.8081	-20.3323	-29.4055	-61.3092	-129.8550
s_mix, kj/kmol/K	-908.4859				

Now we have the following values for entropy and enthalpy.

$$h_i = -124706.6978 \frac{\text{kJ}}{\text{kmol}} \quad s_i = -908.4859 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}$$

Given that the turbine is 100 percent efficient, we know that the inlet entropy is equivalent to the outlet entropy. Now, we should devise a way to guess a temperature that satisfies this statement.

First, let us again make a guess. This time, we are guessing the outlet temperature to be a value slightly lower than the inlet. Let us guess -10 degrees C. Using the known value of pressure and assuming the mole fraction remains constant, we can perform a flash calculation with the goal of setting the outlet entropy equal to the inlet entropy.

Let us start by making guesses for the K-values. We do this by calculating saturated pressure values. If we are not under supercritical conditions, we can use Antoine's equation, stored in the function **Psat**. If we ARE under supercritical conditions, we calculate the saturated pressure via Wilson's equation, which is stored in the function **WilsonPsat**. With our guess of negative ten degrees, the following values are obtained in units of bar.

N2	CH4	C2H6	C4H10
621.9160	204.7629	17.3063	0.6790

In this case, nitrogen and methane were indeed under supercritical conditions. Next, we evaluate our guess for K with the equation below.

$$K_i^0 = \frac{P_i^{\text{sat}}}{P}$$

The values obtained are shown below.

N2	CH4	C2H6	C4H10
124.3832	40.9526	3.4613	0.1358

Now that we have K-values, we can proceed by varying the vapor fraction to solve for the Rachford-Rice equation, as shown below.

$$\sum_i \frac{z_i (K_i^0 - 1)}{1 + \alpha (K_i^0 - 1)} = 0$$

Using the resulting vapor fraction allows to calculate values for the liquid and vapor fractions. We do this by using the equations below.

$$x_i = \frac{z_i}{1 + \alpha (K_i^0 - 1)} \quad y_i = K_i^0 x_i$$

Then, as we have done many times throughout this report, we again calculate compressibility factors and resulting fugacity coefficients. This allows us to calculate new K values, which we compare to our initial guess. If they are not equal, we repeat the procedure until they do become equal. Again, there is a shortcut to do this by using the **Calculate Vapor Fraction** command in the Excel ribbon.

Now, before we proceed in calculating the vapor fraction, let us set up a method of calculating the outlet entropy. It is the same as before, however we must account for the fact that the mixture is two-phase. So, we extend the table to calculate residual values of entropy based on the different compressibility factors. When summing them up, we also consider the vapor and molar fractions. This creates a second table.

	N2	CH4	C2H6	C4H10	sum
Sig, kj/kmol/K	-15.8285	-96.9281	-191.4278	-387.3980	-691.5825
SR, kj/kmol/K	-39.2697	-40.7517	-49.5098	-79.0771	-208.6083
s_mix, kj/kmol/K	-866.4480				
SR_vap	-0.2042	-0.2361	-0.4349	-1.0048	sum
SR_liq	-39.5888	-41.1412	-50.4298	-84.1784	
SR_mix	-21.6753	-19.8381	-49.2681	-84.0840	-174.8656

The resulting overall value of entropy is highlighted in blue. We use goal seek to alter the temperature until this value is equal to the inlet entropy. If it is, we perform a flash calculation to see if it still holds equal. So, we manually iterate until we find a value of the outlet temperature that satisfies both the Rachford-Rice equation as well as the entropy balance.

Eventually, the temperature somewhat converges to a value of -19.04312 degrees C. This proves to be a weak method, as it rarely converges and is not automated. Nevertheless, the vapor fraction is also given at this point, since the method relies on satisfying the Rachford-Rice equation. It is equal to 0.2725.

Now, the last step to solving this problem to get the work is calculating the outlet enthalpy. Similar to the entropy, we have to consider the two phase outlet. Repeating the process with the respective functions for enthalpy, we get the following table.

	N2	CH4	C2H6	C4H10	sum
HIG, kj/kmol/K	-729.7475	-75773.2019	-86011.4463	-128607.2463	-291121.6420
HR, kj/kmol/K	-0.4676	-0.4989	-0.6920	-1.2455	-2.9040
h_mix, kj/kmol/K	-292885.1818				
HR_vap	-309.4401	-350.0319	-519.7796	-1025.0171	sum
HR_liq	-131.8565	-168.9921	-324.3468	-787.1559	
HR_mix	-209.7961	-318.0964	-434.6977	-800.9498	-1763.5399

Then, the work becomes outlet enthalpy minus inlet.

$$W = h_o - h_i = (-292885.1818) - (-124606.6978) = -167278.484 \frac{\text{kJ}}{\text{kmol}}$$

We have to get this value into the desired units.

$$\hat{h} = -167278.484 \text{ kJ/kmol}$$

First, compute the mixture molecular weight using the mole fractions:

$$\bar{M} = \sum_i z_i M_i$$

$$\bar{M} = 0.15(28.0) + 0.30(16.0) + 0.30(30.1) + 0.25(58.1) \approx 32.6 \text{ kg/kmol}$$

Now convert from kJ/kmol to kJ/kg:

$$h = \frac{\hat{h}}{\bar{M}} = \frac{-167278.484 \text{ kJ/kmol}}{32.6 \text{ kg/kmol}} \approx -5.14 \times 10^3 \text{ kJ/kg}$$

$$h \approx -5.1 \times 10^3 \text{ kJ/kg}$$

Now, we can summarize our answers for problem C.

- Inlet temperature = $\boxed{-5.8730^\circ\text{C}}$
- Outlet temperature = $\boxed{-19.04312^\circ\text{C}}$
- Outlet vapor fraction = $\boxed{0.2725}$
- Power generated = $\boxed{-5.1 \times 10^3 \frac{\text{kJ}}{\text{kg}}}$